Problem Set 2

Sagar Lee Stockwell Clemensen (FCK916)

Advanced Methods in Applied Statistics

25th of February 2026

Problem 1

Problem 1a

Using an unbinned likelihood fit from iMinuit the best-fit values of $\mu, \sigma,$ and ξ are extracted. iMinuit assumes Gaussian uncertainties so in order to properly extract the uncertainties a parametric bootstrap is performed. The distributions of the values are shown in fig. 1. The confidence interval values are then taken as the best-fit values $\pm$ uncertainty. The confidence interval is calculated such that the integral from the lower value of the interval to the mean is the same as from mean to the upper value of the interval. The best-fit values and uncertainties are listed in table 1

Variable	μ	σ	ξ
Value	2960^{+52}_{-43}	614^{+29}_{-36}	$-0.22^{+0.04}_{-0.05}$

Table 1: Best-fit values of $\mu, \sigma,$ and ξ with uncertainties

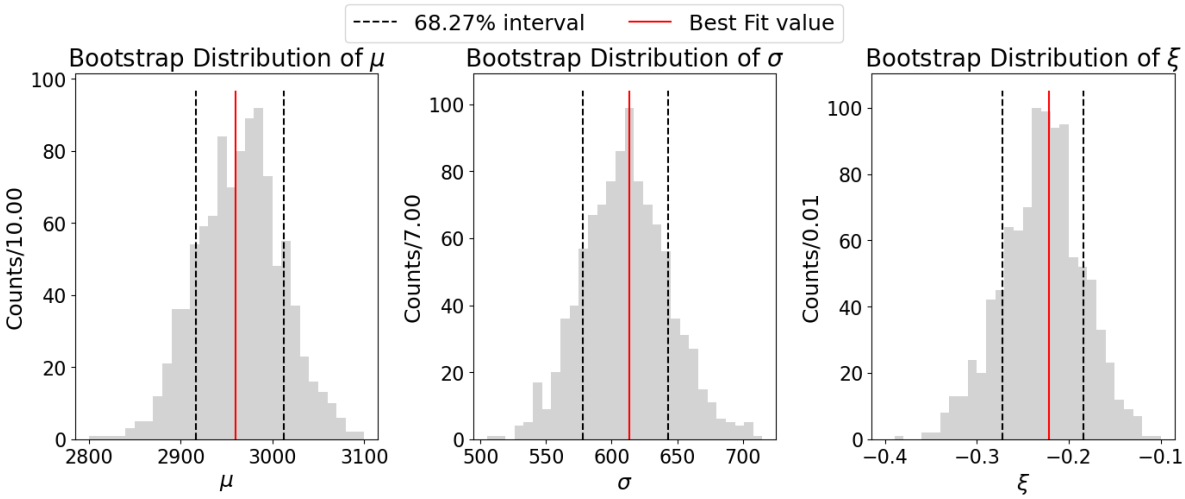

Figure 1: Distributions of parametric bootstrap values of $\mu, \sigma,$ and ξ and 68.27% confidence interval.

A histogram of the 10k data, and a scaled GEV function plotted using the best-fit values is shown in fig. 2. The negative ln-likelihood (-LLH) of the data using the extracted best-fit values is listed in the legend of the plot.

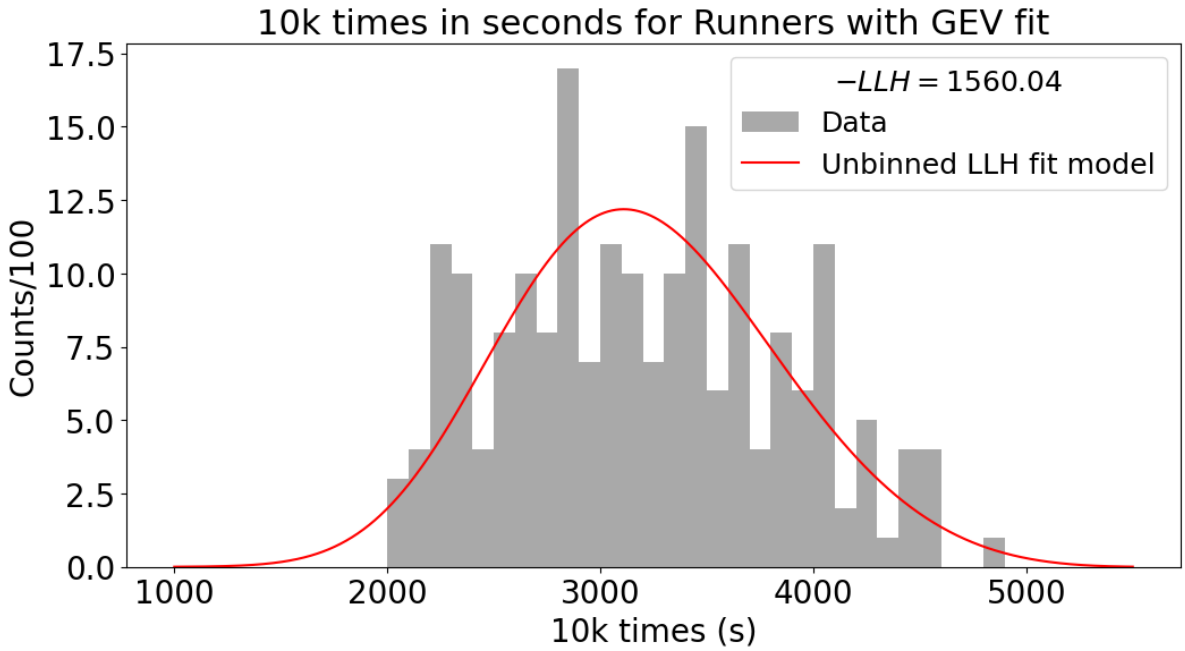

Figure 2: Histogram of the 10k data along scaled GEV function plotted using the obtained best-fit values. The -LLH is listed in the legend.

Problem 1b

Since a CDF exists for the GEV distribution, it can be used to calculate the probability of obtaining a finishing time slower than that of "Tanyusha, SomethingAboutGoats". This probability can also be computed numerically. The analytical and numerical results are identical. Therefore, only the analytical result is reported.

Table 2 lists both the fraction of runners in the dataset who ran slower than "Tanyusha, SomethingAboutGoats" and the corresponding probability calculated from the analytical CDF. The percentage error, defined as

$$\frac{p_{\text{analytical}} - p_{\text{observed}}}{p_{\text{observed}}} \times 100\%, \tag{1}$$

is -12.3%. This is sufficiently large to where I would argue that there is disagreement between the model and the data.

However, if $\mu + \sigma_\mu$ and $\xi + \sigma_\xi$ are used instead of the best-fit parameter values, the analytical probability of running slower becomes 0.228, which is almost exactly the same the observed fraction. Since these parameter values lie within one standard deviation (not to be confused with the GEV scale parameter σ) of the best-fit estimates, this indicates that the model can produce probabilities that are in reasonable agreement with the data within parameter uncertainty even though it doesn't do so for the best-fit values.

Category	Fraction from actual data	Analytical	Percentage error
Value	0.227	0.199	-12.3%

Table 2: The probability from the GEV PDF of running a time slower than that of "Tanyusha, SomethingAboutGoats", the fraction of runners that actually ran slower and the percentage error of the analytical from the actual fraction.

Problem 1c

Using the given values of $(\mu, \sigma, \xi) = (2980, 607.5, -0.26)$ the probability from the GEV distribution of running faster than 2005 seconds, the fastest time in the data set, is 0.0218 ie.

$$P(X < 2005) = 0.0218. \tag{2}$$

A plot of the scaled version of the CDF of the GEV function along with the histogram of the 10k data is shown in fig. 3.

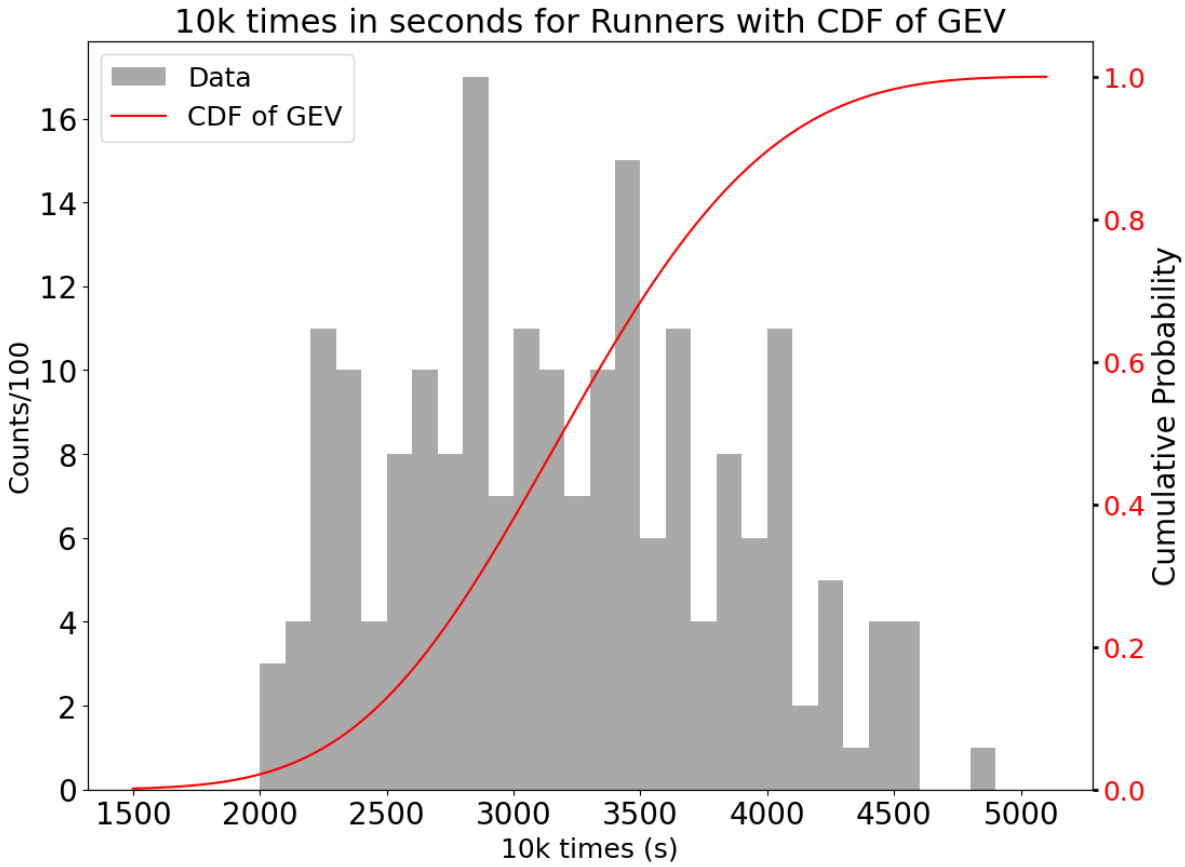

Figure 3: Scaled version of the CDF of the GEV function using $(\mu, \sigma, \xi) = (2980, 607.5, -0.26)$, along with a histogram of the 10k data.

Problem 2

A plot of the likelihood, prior, and bayesian posterior over the range of $0 < x < 4$ is shown in fig. 4. By solving the equation $\frac{1}{N} \cdot 1.6 + \frac{2.66}{N} \cdot 0.5 = 1$ for N, we get $N = 2.93$. Using the bayesian posterior $\hat{x} = 0.5$ and $g(\hat{x}) = 0.683$.

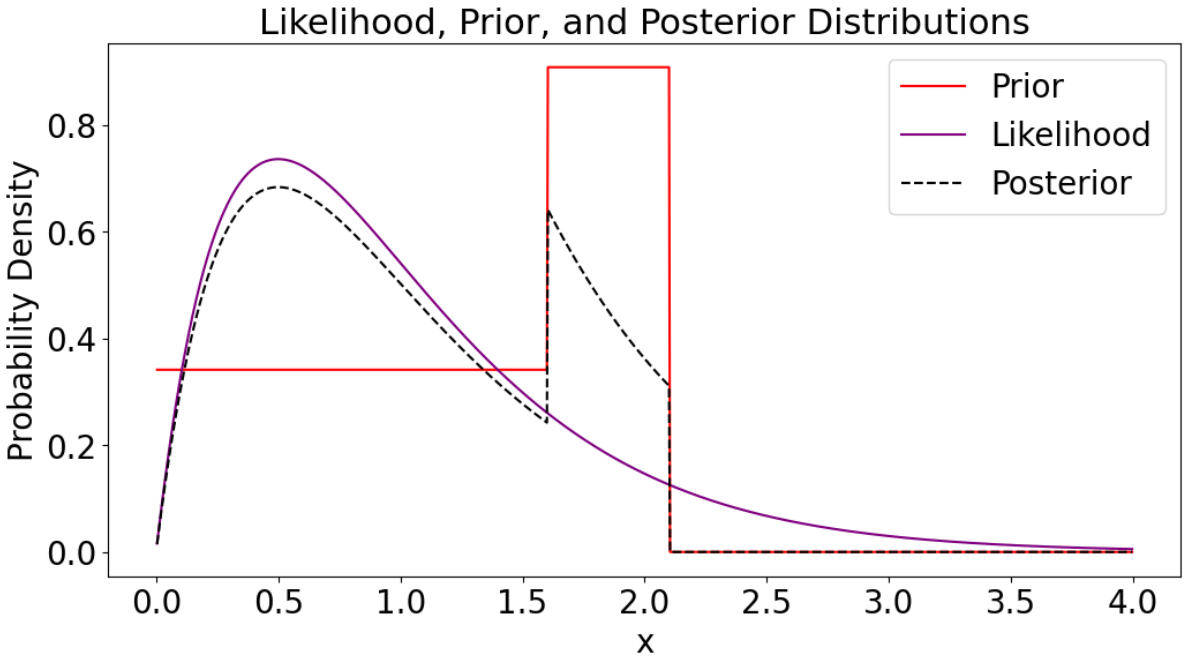

Figure 4: Normalized plots of the likelihood, prior, and bayesian posterior

Problem 3

Problem 3a

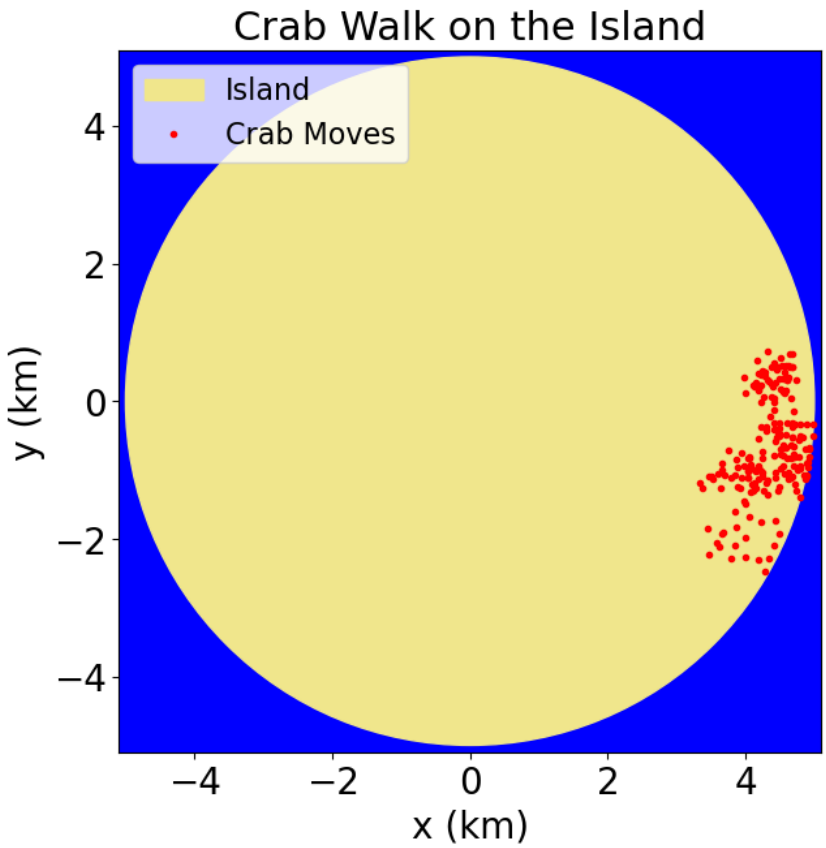

Figure 5: Movement of a single crab starting at (3.6 km, −2.0 km) and moving for 200 days. (It looks like like some of the points are crossing the circle boundary, but they are actually just on the boundary. This is double checked in my code. The crossing is due to the marker size, which has purposely been chosen to be this size, so that the points are visible).

Problem 3b

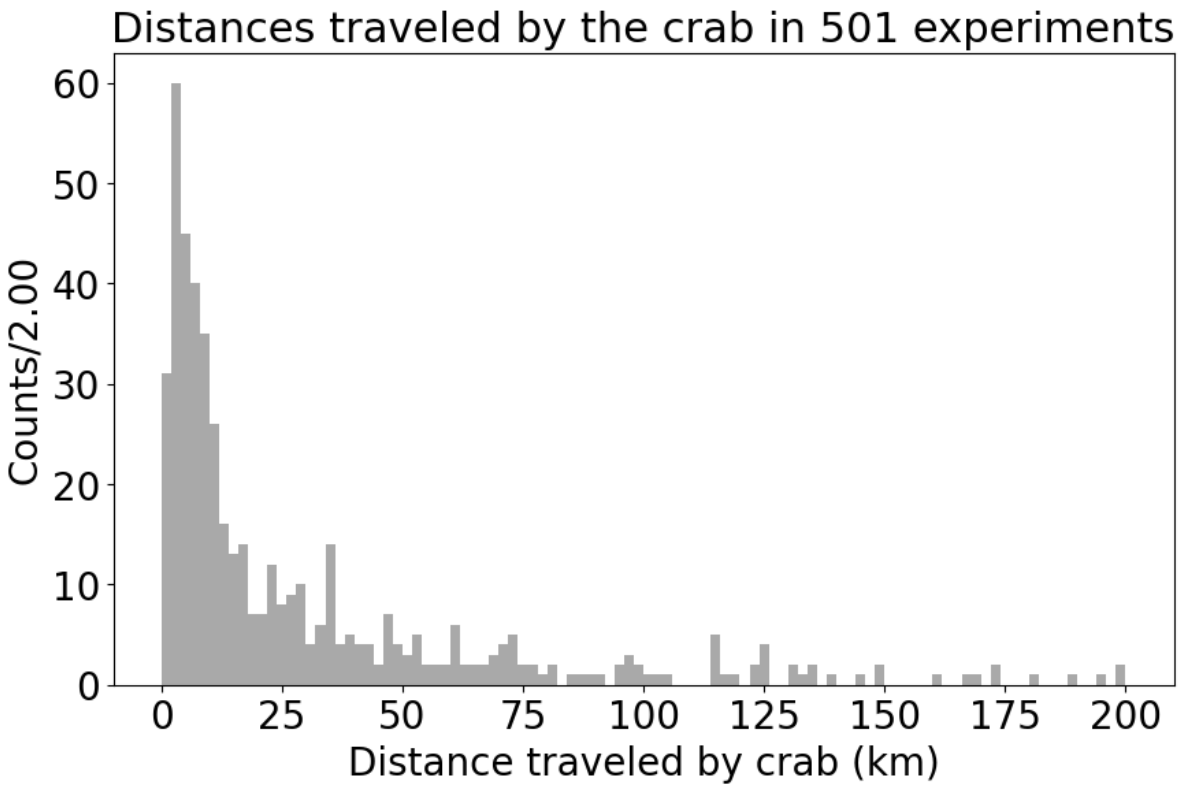

Figure 6: Distance traveled by a signle crab starting at $(3.6\text{ km}, -2.0\text{ km})$ and moving until it arrives at the edge of the island.

Problem 3c

The distribution of the number of crabs remaining alive after 200 days based on 10000 experiments is shown in fig. 7 while the distribution of the mass of the largest crab is shown in fig. 8.

Most likely number of individual crabs that remain alive	Most likely mass of the largest crab
13	3 kg

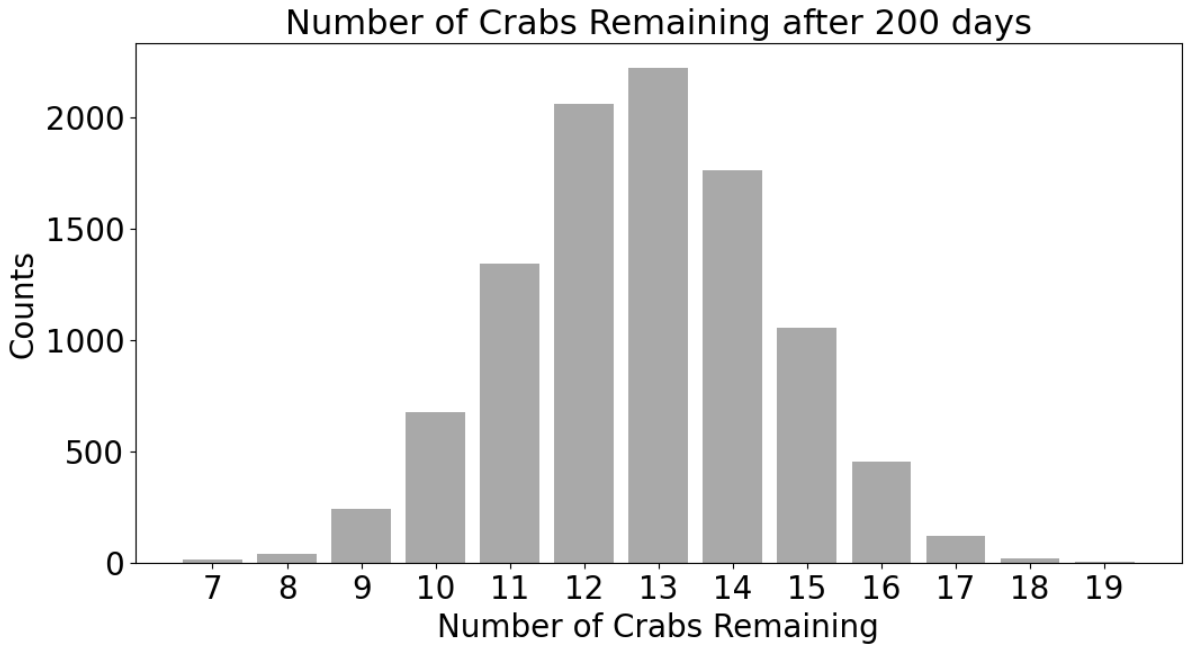

Figure 7: Distribution of the number of crabs remaining alive after 200 days based on 10000 experiments.

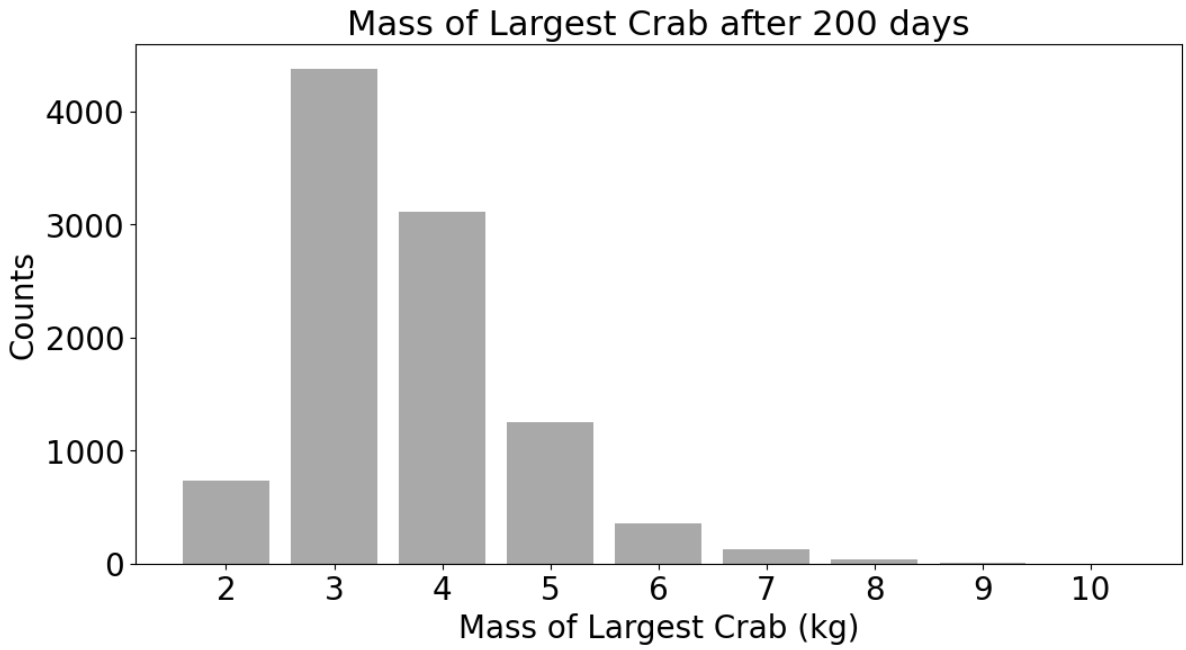

Figure 8: Distribution of the mass of the largest crab after 200 days based on 10000 experiments.

Problem 3d

The distribution of days it takes until only 10 crabs remain alive is shown in fig. 9. The 68.27% confidence interval of the number of days at which only 10 crabs remain is

$$\mathrm{CI} = [222, 425]. \tag{3}$$

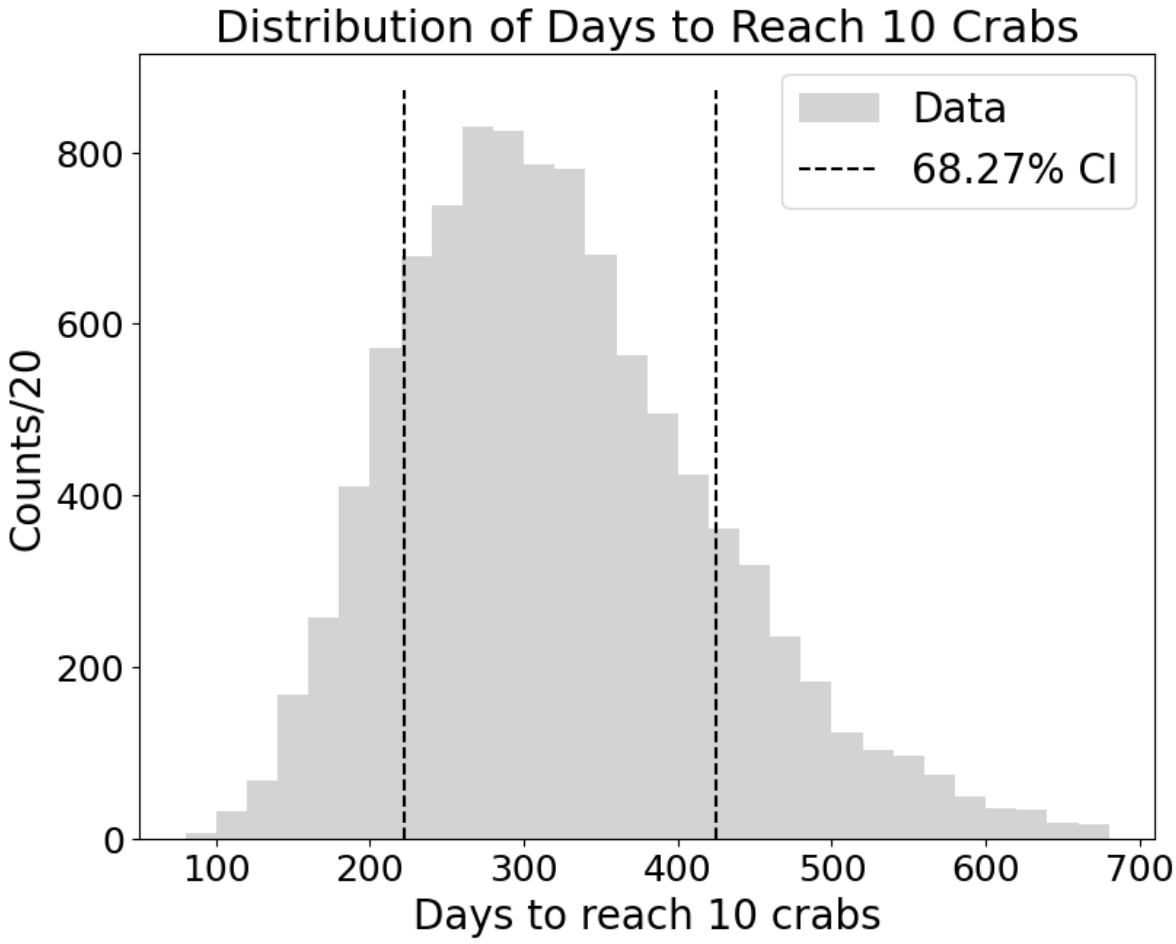

Figure 9: Distribution of days it takes until only 10 crabs remain alive based on 10000 experiments. The 68.27% confidence interval is also plotted.